

Instructions for Lab n

Instructions

Follow these steps to complete the DSA lab efficiently. Make sure to read and understand each step before proceeding.

Step 1: Extract the .zip file

Unzip the downloaded zip file. Only start editing the files after unzipping.

Step 2: Read the lab questions

Refer to the lab questions provided in the file `Q1.pdf` (and `Q2.pdf`, if available).

Step 3: Choose your preferred language

If preferred language is C++, skip this step. If it is `python`, modify the `language.ini` file and set `language = PYTHON` (PYTHON in all caps).

The binaries which make up the autograder will ignore files with other extensions (i.e. `.py` files if `language = C++` and vice-versa), so ensure that you modify the correct solution file while attempting problems.

Step 4: Grant execute permissions to files

Run the following command to grant execute permissions to the required files:

```
:~$ chmod +x RunTestCase CreateSubmission
```

Step 5: View input and expected output for test cases

To view the input and expected output for test case T1, use the following command:

```
:~$ ./RunTestCase Ln Q1 YourBITSId T1
```

Replace `YourBITSId` with your 13-character BITS ID, written in uppercase. Replace `Ln` with the lab number (e.g., L1, L2, L3, etc.).

Step 6: Compile and run your program

Modify the file `Ln_Q1_solution.cpp` (or `Ln_Q1_solution.py`, per preferred language) to solve problem Q1. Use the following command to compile and run your program for test case T1:

```
:~$ ./RunTestCase Ln Q1 YourBITSId T1
```

Repeat this step until your program passes all test cases (e.g., T1, T2, etc.; there are usually 4 of these).

Step 7: Run evaluative test cases

Run the evaluative test cases using the following command:

```
:~$ ./RunTestCase Ln Q1 YourBITSId ET1
```

Repeat this step until your program passes all evaluative test cases (e.g., ET1, ET2, etc.; there are usually 4 of these).

Step 8: Solve question 2 (if applicable)

If the file Q2.pdf is present, follow Steps 5 and 6 for Q2. Modify the file Ln_Q2_solution.cpp (or Ln_Q2_solution.py) accordingly. Replace Q1 with Q2 in the commands.

Step 9: Create and submit the .zip file

When you are ready to submit, create the zip file using the following command:

```
:~$ ./CreateSubmission Ln YourBITSId
```

The output of the above command should show that all the C++ (or python) files have been included in the zip file. Upload the zip file to Quanta (<https://quanta.bits-goa.ac.in>). Do **not** rename or modify the zip file in any manner. Only zip files created using the CreateSubmission script will be evaluated. Files created by other methods will **not** be evaluated.

Important Note

The lab ends at **2:50 PM**, but the RunTestCase executable will only allow you to run visible testcases (T1, T2, etc.) until **2:45 PM**. In these last 5 minutes, you are expected to use your solution in its current state to create the final submission, and to upload the same to Quanta.

Submissions will only be accepted through Quanta and must be submitted before the **2:50 PM** deadline. Submissions via email or other means will **not** be accepted. Ensure your submission follows the instructions to avoid disqualification.